Thinking With Videos: Multimodal Tool-Augmented Reinforcement Learning for Long Video Reasoning

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Figure 1. **Comparison between text-based CoT (left) and multimodal CoT (right) on temporal grounding task.** Green text denotes correct inference and orange text denotes wrong inference. “Thinking with tools” reduces hallucination in the reasoning process by integrating relevant, densely sampled video clip frames into multimodal CoT, resulting in more accurate grounding.

Abstract

*The video reasoning ability of multimodal large language models (MLLMs) is crucial for downstream tasks like video question answering and temporal grounding. While recent approaches have explored text-based chain-of-thought (CoT) reasoning for MLLMs, these methods often suffer from limited cross-modal interaction and increased hallucination, especially with longer videos or reasoning chains. To address these challenges, we propose Video **I**ntelligence via **T**ool-**A**ugmented **L**earning (**VITAL**), a novel end-to-end agentic video reasoning framework. With a visual toolbox, the model can densely sample new video frames on demand and generate multimodal CoT for precise long video reasoning. We observe that temporal grounding and question answering are mutually beneficial for video understanding tasks. Therefore, we construct two high-quality multi-task video reasoning datasets MTVR-CoT-72k for supervised fine-tuning and MTVR-RL-110k for reinforcement learning. Moreover, we propose a Difficulty-aware Group Relative Policy Optimization algorithm (DGRPO) to mitigate difficulty imbalance in multi-task reinforcement learning. Extensive experiments on 14 challenging video understanding benchmarks demonstrate the advanced reasoning ability of VITAL, outperforming existing methods in video question answering and temporal grounding tasks, especially in long video scenarios. Code is available at the project page.*

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1. Introduction

Video understanding is a fundamental challenge in artificial intelligence, with wide-ranging applications such as recommendation [30], smart surveillance [5], and autonomous driving [67]. Recent advances in multimodal large language models (MLLMs) [1, 10, 25, 32, 62] have boosted the ability to jointly process visual and textual information, opening new opportunities for complex video reasoning tasks.

Video reasoning refers to inferring objects, events and their relationships from video content, requiring multi-step and temporal understanding [84, 93]. This capability is fundamental for downstream tasks like video question answering [14], temporal grounding [15], spatial-temporal grounding [16] and captioning [3, 54], where accurate reasoning over dynamic scenes enables more precise and informative results. Inspired by the success of DeepSeek-R1 [17] in enhancing the reasoning ability of LLMs through Group Relative Policy Optimization (GRPO), some works apply GRPO post-training to improve MLLMs' reasoning abilities on images [2, 43, 53] and videos [13, 38].

Despite these advances, most existing MLLMs rely solely on **text-based chain-of-thought** (CoT) reasoning for video understanding [11, 13, 38, 56]. While effective in some scenarios, such approaches often suffer from two major limitations: (1) insufficient cross-modal interaction, which restricts the model's ability to fully leverage visual information during reasoning, and (2) increased hallucination, especially when handling long videos or long reasoning chains. Therefore, a key question arises: *How can MLLMs perform effective reasoning over long videos, with strong cross-modal interaction and minimal hallucination?*

To address these challenges, we propose evolving from text-based CoT to **multimodal CoT** reasoning, where the model can explicitly incorporate visual tools and dynamically attend to relevant video content throughout the reasoning process (Fig. 1). Our **multimodal CoT** reasoning explicitly enhances cross-modal interaction and dynamically grounds predictions in visual content, effectively overcoming insufficient interaction and hallucination issues. This improved paradigm enables more accurate and interpretable video understanding (Fig. 2), highlighting the effectiveness of our approach and confirming that integrating multimodal CoT is a key factor in advancing long video understanding.

Motivated by this insight, we propose **V**ideo **I**ntelligence via **T**ool-**A**ugmented **L**earning (**VITAL**), a novel end-to-end agentic framework for video multimodal (CoT) reasoning. It enables efficient multi-round multimodal tool-augmented training and evaluation. As shown in Fig. 3, VITAL consists of a MLLM and a visual toolbox, which allows the model to actively sample new frames and extract relevant multimodal information on demand during reasoning. This enables the model to focus its attention on critical temporal segments, effectively bridging the gap

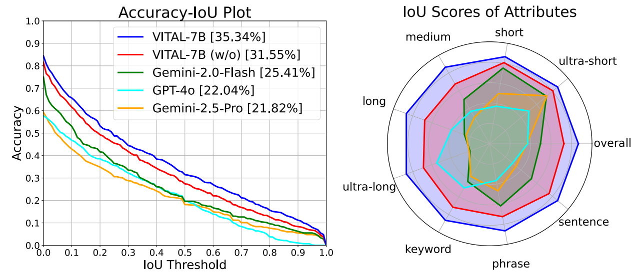

Figure 2. **Performance on long video temporal grounding benchmark VUE-TR.** VITAL-7B achieves state-of-the-art.

between textual reasoning and visual evidence and thus reducing hallucination. To enhance the model's multimodal reasoning abilities for efficient tool calling, we first conduct a cold-start phase, followed by reinforcement learning. In both phases, the model is jointly optimized on temporal grounding, video question answering, and grounded question answering tasks. To support this training paradigm, we construct two large-scale, high-quality training datasets tailored for multi-task video reasoning: **MTVR-CoT-72k** for supervised fine-tuning (SFT) and **MTVR-RL-110k** for reinforcement learning (RL), as shown in Fig. 5. Moreover, we propose a novel Difficulty-aware Group Relative Policy Optimization (**DGRPO**) algorithm to mitigate the difficulty imbalance in multi-task reinforcement learning. By adjusting the reward scale based on task difficulty and sample difficulty, DGRPO ensures adaptive difficulty balancing, leading to more stable training and improved generalization ability. Our contributions include:

- We design an end-to-end MLLM framework that seamlessly integrates video tool use and multimodal reasoning, which generates multimodal CoTs by sampling new frames on demand with a visual toolbox.
- We construct two large-scale, high-quality multi-task video reasoning datasets: MTVR-CoT-72k and MTVR-RL-110k for comprehensive video reasoning learning.
- We introduce a Difficulty-aware Group Relative Policy Optimization algorithm to address difficulty imbalance in multi-task reinforcement learning.
- Extensive experiments on 11 benchmarks and ablation studies demonstrate that VITAL achieves superior performance in long video understanding, video question answering and temporal grounding tasks.

2. Related Works

2.1. Reasoning-Enhanced MLLMs

Recently, OpenAI-o1 [26] and DeepSeek-R1 [17] demonstrate that RL-based post-training can further enhance the reasoning abilities of LLMs. Following the rule-based outcome reward modeling and Group Relative Policy Optimization (GRPO) of DeepSeek-R1, some works apply similar post-training paradigms to MLLMs to enhance multimodal reasoning ability in many tasks like: mathematical and scientific image VQA [24, 49]; image segmentation and grounding [2, 42, 43, 53]; video reasoning VQA [13, 37, 64]; video temporal grounding [38, 48, 69]. **Different from** prior works that rely on text-based CoT reasoning, we leverage multimodal CoT reasoning to effectively improve video understanding ability of MLLMs.

Some works explore the post-training stability of the GRPO algorithm: DAPO [85] proposes a sample-level resampling method; DisCO [35] designs a discriminative objective instead of the GRPO objective. **Different from** them, our DGRPO leverages sample-level re-weighting to efficiently mitigate the difficulty imbalance of multitasks.

2.2. Tool-Augmented LLMs

Recent advances in large language models (LLMs) [61, 81] have shown that equipping models with external tools can enhance their capabilities beyond pure text understanding and generation, and learn to interact with the world. For example, using visual tools can introduce vision ability to LLMs [74, 83]. Based on DeepSeek-R1 [17] style post-training, many works explore using code execution tools for mathematical and coding tasks [12, 34, 39, 66]. Some works leverage search engines as tools for deep search [27] and deep research [91]. In the multimodal domain, several works apply visual foundation models [41], image editing tools [22], or spatial grounding tools [36, 72, 77] to enhance visual reasoning ability. FAST [59] and MVoT [33] incorporate visual evidence in the reasoning process to form a multimodal CoT for image tasks. Inspired by DeepSeek-R1 [17] and OpenAI-o1 and o3 [26], recent works DeepEyes [92] and OpenThinkImg [58] explore "thinking with images" capability by integrating image zoom-in, sketching, detection and segmentation tools to MLLMs for image reasoning tasks. **Different from** prior methods, we focus on enhancing "thinking with videos" capability for long video reasoning by leveraging video grounding and clipping tools.

2.3. Long Video Understanding

Understanding long videos poses significant challenges for MLLMs due to the high computational complexity. Earlier works focus on object-centric feature extraction [75, 89]. Some MLLM works [57, 73], LLaMA-VID [40] and Flash-VStream [87] propose MLLM visual token compression techniques for long videos. LongVA [88] and LongVILA [9] propose long context extension finetuning, which enables training with thousands of frames per video. However, these approaches tend to be computationally intensive. Recent works [20, 46, 68, 70, 71] explore the course-to-fine dynamic frame sampling strategy for efficient long video understanding. **Different from** previous works, we propose a multimodal tool-augmented RL framework for efficient and accurate long video reasoning.

3. Method

We propose VITAL, an end-to-end agentic video reasoning framework that enables "thinking with videos" by generating multimodal chain-of-thought (CoT) using a visual toolbox. As depicted in Fig. 3, VITAL follows the Visual Encoder-LLM architecture commonly used in MLLMs, and integrates a tool-augmented multi-round learning process with the Difficulty-aware GRPO algorithm. In the following subsections, we first introduce the tool-augmented learning framework of VITAL (Sec. 3.1), then discuss the preparation of the multi-task video reasoning training data (Sec. 3.2), and finally present the Difficulty-aware GRPO algorithm (Sec. 3.3), which is employed to optimize the proposed framework with the constructed training data.

3.1. Tool-Augmented Learning Framework

The tool-augmented learning framework operates through a multi-round generation process. In each round, the MLLM determines whether to invoke a tool from the visual toolbox.

Multi-round Generation. Given a user question $\mathcal{T}_0$ and a video $\mathcal{V}_0$ as input, the VITAL model learns to reason and dynamically decides whether to call a visual tool or directly output an answer. If a tool is called, the next round begins after the tool execution, forming a multimodal CoT. At round-k, the model generates:

$$\mathcal{O}_k = f_{\text{MLLM}}\big(\{\mathcal{T}_i, \mathcal{C}_i, \mathcal{V}_i\}_{i=0}^{k}\big) \tag{1}$$

After round-k generation, the parser extracts the thinking step and tool call $\mathcal{T}_{k+1}, \mathcal{C}_{k+1} = p(\mathcal{O}_k)$ or error message $\mathcal{E}_k^p$. Here, $\mathcal{T}_{k+1}$ is the text reasoning step and $\mathcal{C}_{k+1}$ denotes the tool response. ($\mathcal{C}_0 = \varnothing$ is empty.) If a tool call is made, the visual toolbox executes it and returns a new video result $\mathcal{V}_{k+1} = g_{\text{tool}}(\mathcal{C}_{k+1})$ or error message $\mathcal{E}_k^g$. f_{MLLM} denotes the MLLM backbone and g_{tool} denotes the visual toolbox (*e.g.*, video clipping tools). Specifically, if the tool response $\mathcal{C}_{k+1}$ is the final answer, no more tool calls will be made and the process will terminate with the final answer $\mathcal{A}_{k+1} := \mathcal{C}_{k+1}$. This results in a multimodal CoT trajectory:

$$\tau = \{\mathcal{T}_1, \mathcal{C}_1, \mathcal{V}_1, \mathcal{T}_2, \mathcal{C}_2, \mathcal{V}_2, \dots, \mathcal{T}_n, \mathcal{A}_n\} \tag{2}$$

Visual Toolbox. We test several video tools in Tab. 7 and identify the "video clipping" tool as the most effective for video temporal grounding and reasoning tasks. As shown in Fig. 3, "video clipping" tool receives two time range parameters and returns a densely sampled frame sequence of the requested range. The video clipping tool operates as:

$$\mathcal{V}_{k+1} = g_{\text{clip}}(\mathcal{V}_0, t_{\text{start}}, t_{\text{end}}) \tag{3}$$

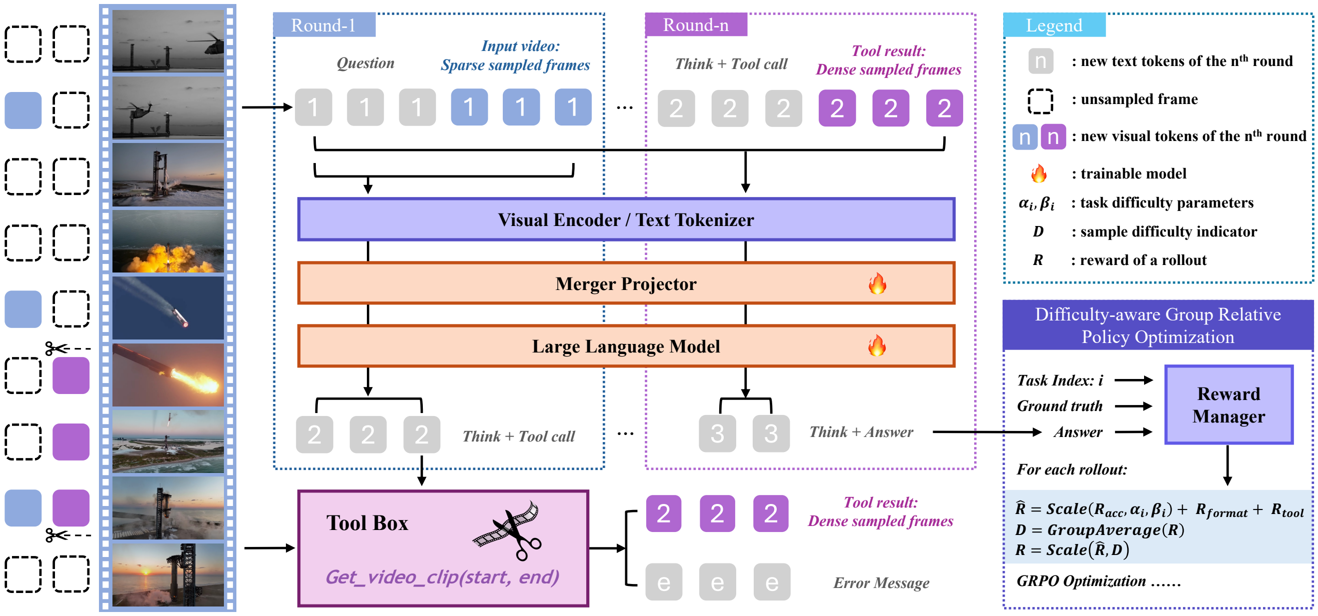

Figure 3. **Overview of the Video Intelligence Tool-Augmented Learning (VITAL) framework.** In the multi-round generation process, the model can attend to video tools adaptively and integrate the tool result to form a multimodal CoT. The model is optimized with Difficulty-aware Group Relative Policy Optimization (DGRPO).

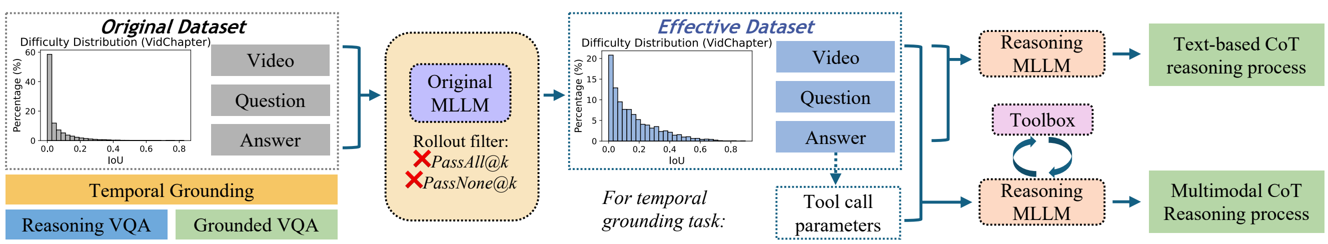

Figure 4. **Data generation pipeline of MTVR training dataset.** A rollout filter is applied to improve data quality.

3.2. Multi-Task Video Reasoning Training Data

To enhance tool calling and multimodal reasoning, we construct two high-quality multi-task video reasoning datasets: MTVR-CoT-72k for SFT cold start and MTVR-RL-110k for RL. These datasets cover video temporal grounding (VTG), video question answering (VQA), and grounded VQA, where temporal grounding serves as the basis for question-guided video clipping, and VQA tasks evaluate general reasoning capabilities. Grounded VQA further requires the model to predict relevant time ranges and answers simultaneously. The original data are collected from Charades-STA [15], ActivityNet-MR [28], VidChapters-7M [80], Video-R1-data [13], LongVideo-Reason [8], ReX-Time [6] and NExT-GQA [78].

As shown in Fig. 4, our data generation pipeline employs a rollout filtering process to improve post-training efficiency. (1) For each sample, we use MLLM [1] to generate k rollouts with a high temperature to encourage diversity. (2) We then retain samples with moderate difficulty, filtering out those where all rollouts pass (*PassAll@k*, too easy) or none pass (*PassNone@k*, too hard). This results in a curated dataset with balanced difficulty. (3) After filtering out extreme data, we leverage a strong reasoning MLLM (*e.g.*, Gemini 2.5 [10]) to roll out the text-based CoT for all data and multimodal CoT for long video data. For long video temporal grounding, we predefine visual tool parameters by adding 20% noise perturbation to the ground truth time range. For long video QA, the reasoning MLLM autonomously selects tool parameters. We use $k = 8$ and temperature $= 1.0$ for both rollout generation and RL training. Filtering criteria for different tasks and CoT generation prompts are detailed in the supplementary material.

Ultimately, we construct four data subsets for multi-stage training: **MTVR-CoT** (54k) and **MTVR-RL** (94k) for basic video reasoning, and **MTVR-CoT-Tool** (18k) and **MTVR-RL-Tool** (16k) for multi-round tool-augmented long video reasoning. Task distribution and data source distribution are demonstrated in Fig. 5.

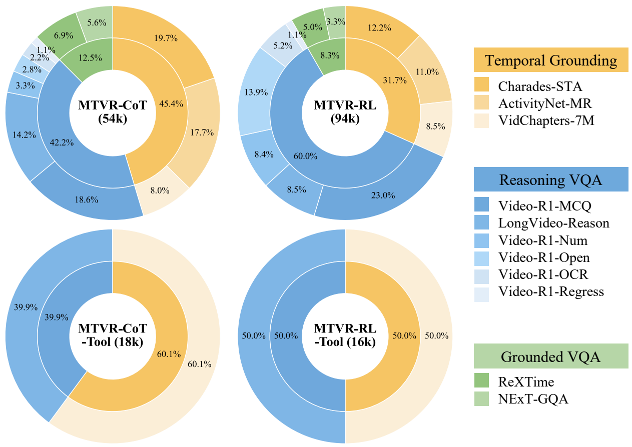

Figure 5. **Task distribution of MTVR training dataset.**

3.3. Difficulty-aware GRPO Training

To address the challenges of multi-task RL training, this section introduces our reward design and difficulty balancing strategies, which ensure stable optimization.

Reward Design. To enable multi-task RL training, we adopt a multi-task *accuracy reward* $\mathcal{R}_{\text{acc}}(\tau)$ following [13]. To keep stable thinking and tool calling, we prompt the model to format its rollout output as:

```
<think> ... </think> <tool_call> {"name":
..., "arguments": ...} </tool_call>
<think> ... </think> <answer> ... </answer>
```

and a strict rule-based *format reward* $\mathcal{R}_{\text{format}}(\tau)$ is applied. In order to encourage the model to attend to the toolbox for new visual information, we add another *tool reward* $\mathcal{R}_{\text{tool}}(\tau)$ to a rollout if it calls at least one tool successfully.

Difficulty Balancing. In our initial explorations, we supervised fine-tuned the model on MTVR-CoT and conducted GRPO training on MTVR-RL with a multi-task objective. However, we noticed there is a phenomenon of difficulty imbalance during GRPO training. For short video datasets and some easy tasks like multiple-choice questions, the reward increases quickly. However, for harder tasks like long video temporal grounding, the IoU reward increases more slowly, which forms *task-wise difficulty imbalance*. We attribute it to the lack of discrimination for continuous IoU function. Another observation is that as the RL training goes on, the proportion of easy samples becomes higher and that of hard samples becomes lower, as discussed in [2], then the optimization soon meets a bottleneck and could not make further breakthrough, which forms *sample-wise difficulty imbalance*. Therefore, we propose a Difficulty-aware GRPO algorithm to address these problems.

Specifically, to mitigate task-wise imbalance, after getting $\mathcal{R}_{\text{acc}}(\tau)$, $\mathcal{R}_{\text{format}}(\tau)$ and $\mathcal{R}_{\text{tool}}(\tau)$ reward of a rollout τ, the accuracy reward (IoU) is scaled conditioned on the difficulty of task-i when the task is temporal grounding (otherwise not scaled), as shown in Alg. 1. Here, α_i, β_i are difficulty parameters of task i. To reduce the sample-wise imbalance, we propose to calculate sample difficulty of task-i, sample-j $D_{i,j}$ by averaging the reward of all G rollouts. Then we apply a soft linear scaling to $D_{i,j}$ to get the difficulty weight of sample-j (ranging from 0.5 to 1).

Algorithm 1 Difficulty-aware Reward Calculation

Require: Trajectories in a batch $\{\tau_{i,j}^k\}$, parameters α_i, β_i
for each task i **do**
  for each sample j **do**
    for each rollout $k = 1, \ldots, G$ **do**
      Compute $\mathcal{R}_{\text{acc}}(\tau_{i,j}^k)$, $\mathcal{R}_{\text{format}}(\tau_{i,j}^k)$, $\mathcal{R}_{\text{tool}}(\tau_{i,j}^k)$
      if task i is temporal grounding **then**
        $S_1 \leftarrow \text{clamp}\left(\frac{\mathcal{R}_{\text{IoU}}(\tau_{i,j}^k)-\alpha_i}{\beta_i-\alpha_i}, 0, 1\right)$
      else
        $S_1 \leftarrow \mathcal{R}_{\text{acc}}(\tau_{i,j}^k)$
      end if
      $\widehat{\mathcal{R}}(\tau_{i,j}^k) \leftarrow S_1 + \mathcal{R}_{\text{format}}(\tau_{i,j}^k) + \mathcal{R}_{\text{tool}}(\tau_{i,j}^k)$
    end for
    $D_{i,j} \leftarrow \frac{1}{G}\sum_{k=1}^{G} \widehat{\mathcal{R}}(\tau_{i,j}^k)$
    $w_{i,j} \leftarrow \text{clamp}(2 - D_{i,j}, 0, 1) \times 0.5 + 0.5$
    for each rollout $k = 1, \ldots, G$ **do**
      $\mathcal{R}(\tau_{i,j}^k) \leftarrow \widehat{\mathcal{R}}(\tau_{i,j}^k) \cdot w_{i,j}$
    end for
  end for
end for
return $\mathcal{R}(\tau_{i,j}^k)$ for all $\tau_{i,j}^k$

After this difficulty-aware reward balancing, we apply the GRPO algorithm based on rollout reward $\mathcal{R}(\tau_{i,j}^k)$ to optimize towards this objective:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{q\sim P(Q),\ \{\tau_k\}_{k=1}^{G}\sim\pi_{\theta_{\text{old}}}(\tau|q)} \left[\frac{1}{G}\sum_{k=1}^{G}\frac{\pi_\theta(\tau_k|q)}{\pi_{\theta_{\text{old}}}(\tau_k|q)}A_k - \beta\,\mathbb{D}_{\text{KL}}(\pi_\theta\|\pi_{\text{ref}})\right] \tag{4}$$

$$\mathbb{D}_{KL}\left(\pi_\theta\|\pi_{ref}\right) = \frac{\pi_{ref}(\tau_k|q)}{\pi_\theta(\tau_k|q)} - \log\frac{\pi_{ref}(\tau_k|q)}{\pi_\theta(\tau_k|q)} - 1 \tag{5}$$

Here $q = \{\mathcal{T}_0, \mathcal{V}_0\}$ denotes the question and input video. As presented in Tab. 5, the DGRPO training is more stable than GRPO and achieves better performance on challenging long video benchmarks LongVideo-Reason (LVR), VidChapters-7M (VidCh), while reserving good accuracy on Video-MMMU and Charades-STA (Cha). Implementation details of DGRPO are included in the supplementary.

Table 1. **Performance on long video temporal grounding benchmarks.** Here $\mathbf{R@x}$ denotes recall at an IoU threshold of x. $\overline{\mathbf{P}}$, $\overline{\mathbf{R}}$, and $\overline{\mathbf{IoU}}$ denote Area Under Curve (AUC) values of precision, recall and Intersection over Union (IoU). Gray rows denote models not open-sourced. VITAL-7B (w/o) denotes VITAL-7B without toolbox. Best results are bold-faced.

Model	Vid-Chapters-7M		
	$\mathbf{R@0.3}$	$\mathbf{R@0.5}$	$\mathbf{R@0.7}$
VTimeLLM-7B [23]	10.6	4.1	1.6
CLIP [51]	10.7	5.2	2.3
M-DETR [31]	37.4	27.3	17.6
ReVisionLLM [20]	33.8	27.4	21.8
Qwen2.5-VL-7B [1]	0.8	0.3	0.1
VITAL-7B (w/o)	35.2	25.8	19.5
VITAL-7B	**45.4**	**34.7**	**24.3**
ΔToolbox	+10.2	+8.9	+4.8

Model	VUE-TR-Vision		
	$\overline{\mathbf{P}}$	$\overline{\mathbf{R}}$	$\overline{\mathbf{IoU}}$
Vidi-1.5-9B [63]	70.0	66.9	49.7
GPT-4o [25]	41.3	30.5	22.0
Gemini-2.0-Flash [10]	48.6	38.9	25.4
Gemini-2.5-Pro [10]	48.2	37.9	21.8
Qwen2.5-VL-7B [1]	44.4	16.0	12.6
VITAL-7B (w/o)	43.7	60.7	31.6
VITAL-7B	**47.3**	**65.0**	**35.3**
ΔToolbox	+3.6	+4.3	+3.7

Table 2. **Performance on long video VQA benchmarks.** VITAL-7B (w/o) denotes VITAL-7B without toolbox. Best results are bold-faced.

Model	Video-MME		LVR	LongVB	LVB	CGB
	Acc	Long	Acc	Acc	Acc	Acc
GPT-4o [25]	71.9	65.3	60.7	66.7	48.9	44.9
Gemini-1.5-Pro [60]	75.0	67.4	69.3	64.0	33.1	37.8
Video-R1-7B [13]	59.3	–	68.1	–	–	–
Video-MTR [79]	59.0	51.0	–	–	–	–
LongVILA-7B [9]	60.1	53.0	62.0	57.1	–	–
LongVILA-R1-7B [8]	65.1	55.2	72.0	58.0	–	–
Qwen2.5-VL [1]	65.2	54.6	60.1	56.0	42.0	44.5
VITAL-7B (w/o)	63.5	53.2	73.2	56.4	43.8	45.0
VITAL-7B	**66.1**	**56.0**	**79.3**	**58.5**	**46.2**	**46.9**
ΔToolbox	+2.6	+2.8	+6.1	+2.1	+2.4	+1.9

Table 3. **Performance on reasoning VQA benchmarks.** VITAL-7B (w/o) denotes VITAL-7B without toolbox.

Model	VSI-Bench	MMMU	MMVU
	Acc	Acc	Acc
LongVA-7B [88]	29.2	23.9	–
VILA-1.5-8B [44]	28.9	20.8	–
LLaVA-OV-7B [32]	32.4	33.8	49.2
AoTD-7B [56]	28.8	–	–
Video-R1-7B [13]	37.1	52.4	64.2
VideoRFT-7B [64]	36.8	51.1	68.5
Qwen2.5-VL-7B [1]	31.8	47.4	61.3
VITAL-7B (w/o)	37.5	52.1	62.6
VITAL-7B	**41.8**	**54.2**	**68.7**
ΔToolbox	+4.3	+2.1	+6.1

4. Experiments

4.1. Experimental Setup

Implementation Details. The VITAL-7B model is implemented with a visual encoder, a merger projector and a large language model pretrained from Qwen2.5-VL-7B [1]. The training framework extends the functionalities of verl [55, 86] and vLLM [29], providing additional support for multimodal tool-augmented training and evaluation with multi-round conversation context.

Training Settings. VITAL is trained with the AdamW optimizer [45] and a cosine learning rate scheduler. The weight decay is 1e-2. The learning rate is 1e-5 for SFT and 1e-6 for RL. The batch size is 256 for SFT and 64 for RL. The number of rollouts is 8 for DGRPO. We train the VITAL-7B model for one epoch at each of the four stages on H100 GPUs, totaling 640 GPU hours.

Evaluation Settings. We evaluated our model on five long video question answering benchmarks: Video-MME [14], LongVideo-Reason (LVR) [8], LongVideoBench (LongVB) [76], LVBench (LVB) [65], CGBench (CGB) [4]; and two long video temporal grounding benchmarks: VidChapters-7M (VidCh) [80] and VUE-TR [63]. We report the results on VUE-TR-Vision to test the visual-language abilities of the models. To evaluate the basic video reasoning and temporal grounding ability of our model, we further compare it with previous works on three complex video reasoning benchmarks: VSI-Bench [82], Video-MMMU (MMMU) [21], MMVU [90]; two video temporal grounding benchmarks: Charades-STA (Cha) [15], ActivityNet-MR [28], and two grounded VQA benchmarks: NExT-GQA [78], ReXTime [6]. We report accuracy (Acc) for VQA tasks, mean Intersection over Union (mIoU) and recalls for temporal grounding tasks. More evaluation details are included in the supplementary.

4.2. Main Results

Long Video Understanding. VITAL-7B achieves state-of-the-art performance on long video temporal grounding and long video question answering benchmarks, as shown in Tabs. 1 and 2. VITAL-7B outperforms previous best open-source models by a large margin on LongVideo-Reason benchmark (Acc: 79.3% vs. 72.0%) and VidChapters-7M (R@0.5: 34.7% vs. 27.4%). The improvement is particularly pronounced in long video scenarios, demonstrating the effectiveness of our tool-augmented multimodal CoT.

Complex Reasoning VQA. As illustrated in Tab. 3, on

Table 4. **Performance on video temporal grounding and grounded VQA benchmarks.** VITAL-7B (w/o) denotes VITAL-7B without toolbox. Best results are bold-faced.

Model	Video Temporal Grounding								Grounded VQA			
	Charades-STA				ActivityNet-MR				NExT-GQA		ReXTime	
	R@0.3	R@0.5	R@0.7	mIoU	R@0.3	R@0.5	R@0.7	mIoU	mIoU	Acc	mIoU	Acc
VTimeLLM-7B [23]	51.0	27.5	11.4	31.2	44.0	27.8	14.3	30.4	28.8	17.4	20.1	36.1
TimeChat-7B [52]	46.7	32.2	15.7	32.2	30.2	16.9	8.2	21.8	14.4	7.6	11.6	40.0
Momentor-7B [50]	42.9	23.0	12.4	29.3	42.6	26.6	11.6	28.5	–	–	–	–
VTG-LLM-7B [19]	52.0	33.8	15.7	–	–	8.3	3.7	12.0	–	–	–	–
TRACE-7B [18]	–	61.7	41.4	41.4	54.0	37.7	24.0	39.0	–	–	–	–
TimeMarker-8B [7]	73.5	51.9	26.9	48.4	67.4	50.7	**33.0**	49.5	–	–	–	–
Time-R1-7B [69]	78.1	60.8	35.3	58.1	58.6	39.0	21.4	40.5	–	–	–	–
VideoChat-R1-7B [38]	82.1	71.7	50.2	60.8	51.8	33.4	17.7	36.6	32.4	70.6	–	–
TimeSearch-7B [47]	73.6	52.4	24.5	48.6	61.0	43.0	26.1	43.9	–	–	36.7	76.5
Temporal-RLT-7B [37]	79.6	67.9	44.1	57.0	56.9	38.4	20.2	39.0	37.3	**78.7**	–	–
DeepVideo-R1-7B [48]	–	71.7	**50.6**	**61.2**	–	33.9	18.0	36.9	36.8	72.5	–	–
Qwen2.5-VL-7B [1]	67.9	50.3	24.3	43.6	28.3	15.8	7.5	21.1	15.4	59.5	27.5	75.7
VITAL-7B (w/o)	81.1	68.1	41.3	57.1	68.5	47.1	26.0	46.6	37.2	77.5	40.9	79.1
VITAL-7B	**83.1**	**72.0**	46.7	59.9	**70.9**	**50.8**	31.6	**49.8**	**43.0**	**78.7**	**47.6**	**80.5**
ΔToolbox	+2.0	+3.9	+5.4	+2.8	+2.4	+3.7	+5.6	+3.2	+5.8	+1.2	+6.7	+1.4

challenging multi-step reasoning benchmarks such as VSI-Bench, Video-MMMU, and MMVU, VITAL-7B consistently outperforms strong baselines, confirming its advanced reasoning capability across diverse video reasoning tasks like spatial reasoning and multi-discipline knowledge learning. For clearer judgement, we test models on MMVU multiple-choice split following previous work [13, 37].

Video Temporal Grounding and Grounded VQA. As shown in Tab. 4, VITAL-7B excels in short video temporal grounding benchmarks (Charades-STA and ActivityNet-MR), indicating that the proposed model has strong basic temporal grounding capability. On grounded VQA benchmarks (NExT-GQA and ReXTime), VITAL-7B also sets new state-of-the-art performance, demonstrating the multimodal CoT design facilitates the integration of accurate temporal grounding and deep video reasoning, leading to more reliable video understanding MLLMs.

4.3. Ablation Study

Effectiveness of DGRPO. We conduct an ablation study on training stages and training styles, *e.g.*, with or without thinking. As shown in row 5 and row 6 of Tab. 5, the introduction of DGRPO improves difficult long video understanding tasks by a large margin while keeping the same short video perception ability, increasing the average score from 50.3 to 52.1. This demonstrates that difficulty-aware reward balancing in DGRPO helps the model better handle diverse task complexities and improves overall robustness.

Analysis of Tool-Augmented Reinforcement Learning. Comparing rows 6 and 7 of Tab. 5, we observe that introducing tool-augmented RL improves long video perception capability, and leads to substantial improvements across all

Table 5. **Ablation study on training stages.** Experiments show that tool-augmented DGRPO is effective for long video reasoning and temporal grounding. * indicates training with tools.

	Train Stage	Style	LVR Acc	VidCh mIoU	MMMU Acc	Cha mIoU	Avg
①	Qwen2.5-VL	No think	60.1	0.5	47.4	43.6	37.9
②	SFT	No think	62.0	10.8	49.9	46.0	42.2
③	SFT+GRPO	No think	63.3	23.5	50.2	56.2	48.3
④	SFT	Think	62.8	15.6	50.5	46.8	43.9
⑤	SFT+GRPO	Think	66.0	25.8	52.0	57.2	50.3
⑥	SFT+DGRPO	Think	70.2	28.8	52.1	57.1	52.1
⑦	SFT+DGRPO +SFT*+DGRPO*	Think	79.3	35.0	54.2	59.9	57.1

benchmarks. The delta results ΔToolbox in the last row of Tabs. 1 to 4 further highlight the consistent benefits of tool integration across diverse video reasoning tasks.

Analysis of Training Dataset Composition. We categorize MTVR-CoT-72k and MTVR-RL-110k by task types, and train VITAL-7B on different task combinations, each for four stages. As shown in Tab. 6, combining temporal grounding (TG), reasoning VQA (RQA), and grounded VQA (GQA) data yields the best overall performance, with the average score improving from 37.9% (without training data) to 57.1%. By learning from diverse objectives, multi-task training builds **general and robust representations**. This inherent advantage allows it to consistently outperform models trained on a single, isolated task. This result highlights that multi-task training with all three data types provides strong synergy and is crucial for robust video reasoning and temporal grounding.

Analysis of Visual Tools. Tab. 7 compares different visual tools on zero-shot temporal grounding tasks. We adopt two

Table 6. **Analysis of training data compositions with the same size.** Combining temporal grounding, reasoning VQA and grounded VQA brings the best result. Rows 2-5 conduct the same four stages training as Exp. ⑦.

	Training Data	Size	LVR Acc	VidCh mIoU	MMMU Acc	Cha mIoU	Avg
①	None	0	60.1	0.5	47.4	43.6	37.9
	TG	73k	59.5	24.8	44.5	52.8	45.4
	RQA	73k	76.4	4.8	49.3	38.9	42.4
	TG+RQA	73k	76.5	31.8	50.1	53.6	53.0
	TG+RQA+GQA	73k	76.3	32.0	51.2	55.9	53.9
⑦	TG+RQA+GQA	182k	79.3	35.0	54.2	59.9	57.1

Table 7. **Analysis of utilities of different visual tools.** We test the zero-shot temporal grounding ability of GPT-4.1 and Gemini-2.5-Pro with different tools. For clip caption and QA tools, the MLLM invokes a sub-agent of the same model.

Style	Tool	Vid-Chapters-7M		Charades-STA	
		GPT	Gemini	GPT	Gemini
Not think	–	26.3	28.1	31.5	32.1
Think	–	28.4	29.8	30.1	33.0
Think	Clip caption	2.0	8.0	30.8	27.8
Think	Clip QA	2.0	10.4	25.5	28.1
Think	Video clip	25.5	26.3	28.8	31.7

Table 8. **Ablation study on VITAL using different visual tools.** Experiments of row 2-4 all start from the checkpoint of Exp. ⑥ and conduct tool-based stage-3/4 training.

	Tool	Tool model	LVR Acc	VidCh mIoU	MMMU Acc	Cha mIoU	Avg
⑥	–	–	70.2	28.8	52.1	57.1	52.1
	Clip caption	Gemini-2.5-Pro	73.5	30.1	49.7	53.9	51.8
	Clip QA	Gemini-2.5-Pro	74.1	32.5	51.3	54.2	53.0
⑦	Video clip	–	79.3	35.0	54.2	59.9	57.1

strong reasoning MLLM GPT-4.1 and Gemini-2.5-Pro as baselines. The evaluation setting is the same as VITAL. We observe that adding clip caption tool or clip QA tool leads to significant mIoU drops for video temporal grounding. We suggest that this is due to the zero-shot setting and the models are not optimized with agentic visual tools. We adopt Gemini-2.5-Pro as the annotator and tool executor.

VITAL is a general and extendable agentic framework. We explore the functionality of the three tools by annotating three versions of MTVR-CoT-Tool (18k) and conducting stage-3/4 SFT and DGRPO for fair comparison. As shown in Tab. 8, video clip tool is the most effective in all tasks, while clip caption / QA tool is slightly lower. We attribute it to the output domain gap of different LLMs and the potential hallucinations caused by QA and captioning. Therefore, we adopt video clip tool by default, since it will not cause textual hallucination and is more efficient. Tool implementation details are in supplementary.

Statistics of Tool Calls. As illustrated in Fig. 6, the VITAL-7B model demonstrates flexible and adaptive tool usage across various tasks like long video VTG and VQA, typically making zero to two tool calls per sample. Notably, the model achieves optimal results with moderate tool usage, indicating its ability to judiciously determine when and how to invoke tools, avoiding overuse or low efficiency. A comprehensive efficiency analysis is in the supplementary.

4.4. Qualitative Analysis

Fig. 1 presents a qualitative comparison between text-based CoT and multimodal CoT on the temporal grounding task. The former relies solely on textual reasoning, which leads to inaccurate self-reflection and a lower IoU score due to error accumulation and hallucination. In contrast, multimodal CoT leverages visual evidence during reasoning, which allows the model to precisely determine the temporal boundaries of the target event. More success and failure case studies are provided in the supplementary material.

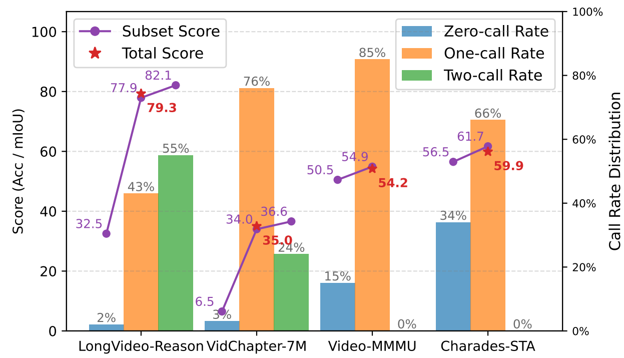

Figure 6. **Tool call statistics on some benchmarks.** VITAL model usually makes zero to two tool calls per sample. This analysis highlights moderate tool usage often brings better results.

5. Conclusion

In this work, we introduce VITAL, a novel tool-augmented framework that empowers MLLMs with advanced long video reasoning capabilities, which effectively mitigates hallucination and enhances cross-modal interaction. By integrating a visual toolbox and enabling multimodal CoT generation, VITAL effectively mitigates hallucination and enhances cross-modal interaction, particularly in challenging long video scenarios. We further construct two high-quality multi-task video reasoning datasets and propose the DGRPO algorithm to address the difficulty imbalance in multi-task reinforcement learning. Extensive experiments on 14 benchmarks demonstrate that VITAL achieves state-of-the-art performance in reasoning VQA and temporal grounding tasks. Our results highlight the importance of tool-augmented multimodal reasoning and provide valuable insights for future research in long video understanding.

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